GMU - FACULTY PROFILE

Contact

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Dr. Kavitha K J

Associate Professor, ECE Department, School of Engineering, GM University
Associate Dean Research, GM University
Davanagere - 577006

Faculty
Faculty of Engineering and Technology

School / Program
School of Engineering
B.Tech. in Electronics and Communication Engineering

Faculty Introduction

I am presently serving as the Associate Professor in ECE Department, School of Engineering and Associate Dean Research at GM University, Davanagere. I have twenty-one years of experience in teaching and research in the field of Electronics and Communication Engineering. My areas of interest include Signal Processing, Medical image Processing and Communication. I am also supervising the Ph.D. scholars under VTU and GM University. I have handled research project under NAIN and currently working on the ICSSR sanctioned project.

Qualifications

Education	University	Institution	Year	%
Ph. D	Sathyabama University	Chennai	2022	-
M.Tech (CSE)	VTU, Belgaum	B.I.E.T, Davanagere	2009	67 %
B.E (ECE)	VTU, Belgaum	TCE, Gadag	2003	60 %

Teaching Experience (21 years)

#	Institution	Designation	from	To	Exp.
5	GM Institute of Technology	Associate Professor	29-8-2022	Till Date	
4	Jain Institute of Technology	Assistant Professor	16-06-2012	27-8-2022	10 -Year 1-month, 11 days
3	PESITM, Shivamogga	Sr. Lecturer	25-01-2012	31-05-2012	4 -month
2	GM Institute of Technology	Lecturer	25-08-2004	30-11-2011	7 - Year 3-month

Industry Experience

NIL

Research Areas

Signal Processing, Medical Image Processing, Communication

Research Scholars Registered and awarded for Ph.D.

Sl. No.	Research Scholar	USN	Year of Registration	University	Status
1	Gnanika I V	R23E01EC003	2023	GMU	Course work Completed
2	Tanuja T	R24E02EC004	2024	GMU	Course work Completed
3	Lavakumar T B	R25E02EC001	2025	GMU	Registered
4	Neetha M	R25E02EC002	2025	GMU	Registered
5	Manjunatha Y	4GM23PEC02	2023	VTU	Course work

Research Funding Details

PI/Co-PI Name	Title	Agency	Sanction Date	Amount	Duration
Kavitha K J	Sucrose Level Estimation	NAIN	2024	2.5 L	1 Year
Kavitha K J	Tech Sakhi	ICSSR	2025	5 L	1 Year

Patents Details

Faculty Name	Patent Title	Patent Date of Filing with Patent No.	Patent Published Date	Patent Granted Date
Kavitha K J	Sucrose Level Estimator	07/04/2025 202541033938	Not Yet	Not Yet
	Artificial Speaking IoT Based "Talking Hand" An Assistive Device for Dumb People in Regional Language Kannada	26/11/2020 202041051443	27/05/2022	Not Yet

Publications Details

Scopus Journal Papers	Please see Annexure -I	13
Other Journal Papers	Please see Annexure -II	22
Conference Papers	Please see Annexure-III	10

Books/Book chapters	Please see Annexure-IV	5
	Total Publication	

Workshop/ Training

Organized	Please see Annexure -V	
Attended	Please see Annexure-VI	

List out the Consultancy Projects undertaken

1	NIL

List out the Guided/involved in Innovation workshops/Training

1	NIL

Have you done any contribution to new product development and setting up start-ups?

1	NIL

List of Training Courses Taught with number of students in each course

1	

Professional Recognition

Name of Professional Recognition	University/Institute	Year	Remarks
Doctoral committee member	VTU, Belagavi	2022 to till date	3 Research scholar, VTU

Professional Membership

1	Indian Society for Technical Education (ISTE), Life Member (Membership No. LM 96041)
2	IEEE, (Membership No. 101307743)

Skill Set

Languages	MATLAB, Python, Embedded, Machine Learning
Simulation tools	Matlab, SCi Lab, TinkerCad

NPTEL/MOOC Course Completed

Course	Course Name	Course completed Date	Remarks
NPTEL	Effective Engineering Teaching In Practice	Jan-Feb 2025	Elite (82 %)

Awards/ Honour's/ Achievements

1	Organizing Committee - National Conference at GM University - (2025)
2	Successfully completed with a consolidated score of 83%in "Higher Education Assessment Test" Faculty Quiz Competition, held on 13thSeptember 2025 organized by the GMU Quizers, GM University - (2025)
3	Successfully completed the Faculty Development Programme through LEAP with an average score of 89% (Grade A) held on 30th August 2025 at GM university. - (2025)
4	Best Performer with A Grade in the Application Software for Enhanced Course Delivery (ASECD) program on "Arduino IDE, Tinkercad" held from 3rd to 5th October 2024. Organized by SDTC, GM University, Davanagere.- (2024)
5	Best Paper Award – Jain Institute of Technology, Davanagere - (2022)
6	Organizing Committee – International conference at JIT - (2022)
7	Organizing Committee – National conference at JIT - (2016)

Funded Students Projects Guided: PG /UG (6)

Sl. No.	Name of the Students	Title of Project	Institute (KSCST, VTU, Others)	Amount	Year
1					
2					

Invited Lectures

1	Lecture on "IOT", at Nuthana College, Davanagere, 2024.
2	Keynote Speaker at FDP held at JIT, Davanagere, 2022 .

Present Role and Responsibility

1	Associate Dean Research, GM University

Annexure – I (Scopus Journal Papers)

11	Medical image watermarking based on novel encoding for EHR and fusion based morphological watershed segmentation algorithm for medical images, Multimedia Tools and Applications 83 (9), 25163-25190, 2024.
10	Patient Monitoring Using 5G, with MIMO-NOMA for mm-Wave Communications in Heterogeneous Networks, Smart Hospitals: 5G, 6G and Moving Beyond Connectivity.
9	Compressed Sensing Reconstruction Algorithms for Medical Images – A Comparison, Smart Hospitals: 5G, 6G and Moving Beyond Connectivity
8	CNN Ensemble Approach for Early detection of Sugarcane Diseases – A Comparison, International Journal of Electronics and Telecommunications, 70 (02), 455-464
7	Medical Image Watermarking techniques A Comparison, IJFANS International Journal of Food and Nutritional Sciences, Vol. 11, No. 7, 2022, UGC Group-II Journal.
6	An analysis of plant diseases using neural network for increasing economic growth of farmers, Tianjin University Science and Technology, Vol. 56, No. 02, 2023, Q2 Journal.
5	"Corona Virus, Its Effects And Solution-A Review", NeuroQuantology, Vol. 20, No. 10, October 2022, Scopus (Q3).
4	"A Watermarking Scheme using Intellectual Encoding-Encryption based Blind Reversible Integer Wavelet-Singular Value Decomposition Transform for Authenticity Detection". Journal of Ambient Intelligence & Human Computing, 12, 7715–7726 (2021), Scopus, Web Of Science - SCIE-Q1, Impact Factor: 7.104. DoI: 10.1007/s12652-020-02496-9
3	"An Efficient Medical Image Watermarking Technique Using Integer Wavelet Transform And Quick/Fast Response Codes", IJISTA (INDERSCIENCE Publications), Vol. 18(3), P.No.: 271-280, Scopus - Q3, (Elsevier),UGC, Impact Factor: 0.9. DOI: 10.1504/IJISTA.2019.099344.
2	"E2-Inviible watermarking for protecting Intellectual rights of medical images and records", International Journal of Advanced Computer Science and Applications (IJACSA), Vol.. 9, No. 10,P.No.: 267-276, Scopus – Q3, WOS(ESCI), Impact Factor: 0.28. DOI: 10.14569/IJACSA.2018.091032.
1	"Joint Digital Water Marking for Medical images for Improving Security", Biomedical & Pharmacology Journal, Vol.11(2), P.No.: 863-870, Scopus & UGC (Q4).

Annexure – II (Other Journal Papers)

13	Automatic cylinder book and leakage detection system, Journal of Tianjin University Science and Technology, Vol. 56, No. 02, 2023, Q2 Journal
12	A Survey on ECG Signal and Denoising using FIR and Wavelets, Gis science journal, Vol. 9, No. 12, 2023.
11	"IS-BRW: Image Security using Blind Reverse Watermarking with Various Attacks", Indian Journal of Science and Technology, Vol.11(32), P.No.: 1-12, WOS (Zoological record).

10	"Improved Alpha blending algorithm for the Authentication of Medical images using LWT & QR Code", International Journal of Imaging and Robotics, Vol. 19, Issue. 3, P.No: 26-34.
9	"A Literature survey on Digital Watermarking on medical image", International Journal of Engineering Research And Management (IJERM), Vol. 03, Issue. 08, P.No.: 18-21.
8	"An Introduction to the Digital Watermarking", International Journal of Advanced Engineering Research and Science (IJAERS), Vol.3, Issue.6, , P.No.: 157-160.
7	"A Brief study on Steganography techniques", International Journal of Advanced Engineering, Management and Science (IJAEMS), Vol. 2, Issue. 4, P.No.: 180-183
6	"A Study on Digital watermarking on Medical Images", International Journal of Digital Image Processing, Vol. 7(9), P.No: 265-267.
5	"Video watermarking using DCT and DWT, a comparison", European Journal of advances in Engineering and Technology (EJAET). Vol. 2 Issue 6, P.No.: 83-87.
4	"Survey on digital water marking on medical images", International Journal Of Advanced Computer Research, ,Vol 3,No. 4,Issue No. 13
3	"Design of online interactive data acquisition and control system for embedded real time application", International Journal of Applied Science and Technology, Vol. 3, Issue. 1
2	A Technical Survey on Early Detection of Melanoma", International Journal of Applied Science & Technology Research Excellence Vol. 3 Issue 1
1	"An automatic speaker recognition system using matlab", World journal of science and technology, Vol. 3, No 4, Issue 13

Annexure-III (Conference Papers)

10	**Quantum Leap in Healthcare Security: The Role of Quantum Watermarking,** March 08-09, 2025 at National Institute of Technology Uttarakhand, India, 3rd International Conference on Power Engineering and Intelligent Systems (PEIS2025)
9	Enhanced Breast Cancer Detection Using an Ensemble Model of Random Forest and LightGBM: A Comparative Analysis, International Conference on Electrical, Electronics and Automation (E2ACON 2025), Dr B R Ambedkar National Institute of Technology Jalandhar, Punjab, 8-9 March 2025.
8	CNN and LSTM Approach for Soil Nutrient analysis for Sugarcane Crop, International Conference on Electrical, Electronics and Automation (E2ACON 2025), Dr B R Ambedkar National Institute of Technology Jalandhar, Punjab, 8-9 March 2025.
7	FPGA Based Neural Face Recognition Systems – A Survey, International Conference on Business Intelligence and Data Analytics, April 12-13, 2025
6	Advancements in Vedic Multiplier Design: A Comprehensive Review, Conference on Sustainable Computing and Intelligent, University of Canberra, Australia, September 9-10 2025.
5	"Fusion Based Morphological Watershed Segmentation Algorithm For Medical Images" , Research Advances in Intelligent Computing, CRC Press, Taylor & Francis Group. Doi: https://doi.org/10.1201/9781003320340
4	Crypto-Encoding Multiple Medical Image Watermarking Technique for Electronic Health Record, August 11-13, 2023, IIT Roorkee, Uttarkhand,

3	"Fusion Based Morphological Watershed Segmentation Algorithm For Medical Images," International Conference on Advanced Network Technologies and Intelligent Computing (ANTIC-2021).), Department of Computer Science, Banaras Hindu University, Varanasi, India. – 221005, December 17-18 2021.
2	"Prime Based Encoding Algorithm to Ensure Integrity of Electronic Health Records," 3rd International Conference on Communication and Computational Technologies (ICCCT 2021), Rajasthan Institute of Engineering and Technology, Jaipur and Rajasthan Technical University Kota in Association with Soft Computing Research Society, February 27-28, 2021.
1	"Implementation of DWM for medical images using IWT and QR code as a watermark", *2017 Conference on Emerging Devices and Smart Systems (ICEDSS)*, Tiruchengode, P.No.: 252-255, Scopus.

Annexure-IV (Books/Book chapters)

Academic year – 2024-25 (August to July)		No. of Publication	
4	Advancements in Vedic Multiplier Design: A Comprehensive Review, Conference on Sustainable Computing and Intelligent, 2025.		
3	Patient Monitoring Using 5G, with MIMO-NOMA for mm-Wave Communications in Heterogeneous Networks, Smart Hospitals: 5G, 6G and Moving Beyond Connectivity.		
2	Compressed Sensing Reconstruction Algorithms for Medical Images – A Comparison, Smart Hospitals: 5G, 6G and Moving Beyond Connectivity		
1	Prime Based Encoding Algorithm to Ensure Integrity of Electronic Health Records, Proceedings of International Conference on Communication and Computational Technologies, Book Subtitle: ICCCT 2021, published on 24/08/2021, Springer, Singapore. DOI: 10.1007/978-981-16-3246-4.		

Annexure – V (Workshop /Training /FDPs/ Industry Visit Convened & Organized)

21	One-Day Awareness Workshop on the Incubation Scheme under Raising and Accelerating MSME Performance (RAMP)", jointly organized by the District Industries Centre, Davanagere, KCTU Bengaluru, and GM University, Davanagere on 9th October 2025.
20	National Conference NCSEEE – 2025 on 21st August 2025.
19	Two days' Workshop on "Drone and CAD design" on 4th and 5th June 2025 under NAIN.
18	one day workshop on "Acer X Microsoft Campus Day" at AV Conference hall, Central Library, GM University on 16th March 2024
17	Two days program on "Drone Technology Demonstration" in the field of agriculture on 22nd and 23rd February 2023 at Hiremegalagere Village.
16	3-days workshop on "Internet of Things" from 15-06-2023 to 17-06-2023 and 29-12-2022 to 31-12-2022.

15	Technical Talk on the seminar topic "Unmanned Aerial Systems (Drone) – A Technology" for faculties and students held on 3rd October 2022.
14	International Conference on Engineering Innovation, 3rd June, 2022, JIT.
13	2-days Workshop on Python & Machine Learning, 22nd and 23rd December 2021, ECE Dept., JIT.
12	A Seminar on Wireless Communication and its applications, 9th December 2021, ECE Dept., JIT.
11	2-days Hands-on Workshop on Embedded System for Beginners, 29th & 30th October 2021, ECE Dept., JIT.
10	Session on Problem Solving and Ideathon workshop, 11th November 2021, ECE Dept., JIT.
9	Infrastructure requirement for Teams and Zoom cloud ecosystem collaboration platform, 8th October 2021, ECE Dept., JIT.
8	How to plan for Start-Up and Legal & Ethical Steps, 26th August 2021, ECE Dept., JIT.
7	Role of Youths in building Swasth Bharath, 23rd July 2021, ECE Dept., JIT.
6	Social Awareness Program on HIV, Cervical & Breast Cancer, 22nd February 2019, ECE Dept., JIT.
5	Autodesk Eagle PCB Design Software, 14th & 15th February 2019, ECE Dept., JIT.
4	Arduino & Wireless Embedded systems, 13th to 15th October, 2016, ECE Dept., JIT.
3	National Conference on Recent Trends in Communication and Computing, 27th February, 2016, ECE Dept., JIT.
2	Plagiarism Issues and Essential Tools for Research Write-ups, April,2015, ECE Dept., JIT.
1	Workshop on MSP 430 processor, 2012, ECE Dpt.,PESIT, Shimogga

Annexure – VI (Workshop/Training attended)

30	FDP on "ICT Tools for Teaching and Learning Process", Under Faculty and Staff Enrichment Center on10th August 2024, GMU
29	3 Days FDP on Arduino, IDE and Tinkercad from 3rd to 5th October 2024 at GM University, Davanagere. Microsoft, SAP and AICTE led FDP on AI Evolution: From Foundations to Generative AI under TechSaksham from 22nd January 2024 to 27th January 2024.
28	AICTE Training and Learning (ATAL) Academy Blended/Hybrid FDP on"Analog and Digital Design Flow for VLSI Chip Application" from 2023-02-20-2023-02-25 to 2023-02-27-2023-03-03 at GM Institute of Technology, Davangere
27	AICTE-ISTE approved Orientation/Refresher Programme on "Recent Trends in Machine Learning & Pattern Recognition", 06/12/2021 to 11/12/2021, GM Institute of Technology, Davanagere.
26	AICTE STTP on "Signal Processing and its applications using Machine Learning", 13/09/2021 to 18/09/2021, University BDT College of Engineering, Davanagere.
25	ATAL Faculty Development Programme on "Machine Learning in Medical Image Processing", 05/09/2021 to 09/09/2021, Katihar Engineering College, Katihar.
24	ATAL Faculty Development Programme on "VLSI -IP DESIGN Approach to SRAM compiler design", 16/08/2021 to 20/08/2021, GM Institute of Technology, Davanagere.

23	ATAL Faculty Development Programme on "Digital Forensics and Investigations", 19/07/2021 to 23/07/2021, K.L.E Technological University, Hubli.
22	ATAL Faculty Development Programme on "Medical Image Processing and 3D Applications", 05/07/2021 to 09/07/2021, University College of Engineering.
21	“Microwave Sensors For Biomedical Applications”, 26/06/2021, Brindavan College Of Engineering, Dwarakanagar, Yelahanka, Bangalore.
19	“The Complete Guide On Review Paper Publication For PhD Scholars”, 20/06/2021, Research Graduate.
18	“Implementation Of National Education Policy (NEP)”, 07/06/2021, The OXFORD College Of Engineering, Bommanahalli,Hosur Road,Bangalore.
17	“Journal Citation Reports (Jcr) Training & Certification Program 2021”,15/062021 & 17/06/2021, Organised by Clarivate,Web Of Science.
16	“CU-SAT:Chandigarh University Student Satellite Program”, By Padmashri Dr. Y S Rajan, Padma Shri Dr. Mylswamy Annadurai & Padma Shri Prof. R M Vasagam, 01/05/2021,Chandigarh University.
15	“Overview Of Teaching Techniques In Engineering Statistics & Linear Algebra”, 6/01/2020 TO 10/01/2020 At VTU Human Resource Development Centre, Centre For PG Studies, Muddenahalli, Chikkaballapur DIST.
14	AICTE Share And Mentor Margadarshan Scheme GUEST LECTURE SERIES ON “Smart Antennas And Electromagnetics”, 20TH December 2018, at JNN College of Engineering.
13	STTP On “CMOS, Mixed Signal And Radio Frequency VLSI Design”, 26/12/2016 To 4/02/2017, (Physical Participation At Remote Centre 30/01/2017 To 04/02/2017, Indian Institute Of Technology,Kharagpur.
12	“Feel Teacher”, January 29th, 30th, 31st, Jit,Davanagere.
11	“Latex For Documentation”, 2013 9th Nov, GMIT in Association With VTU At Davanagere.
10	“Recent Advancements In The Field Of DSP”, 2011 17th-19thfeb (3 Days), RBYMEC,Bellary.
9	Role Of Faculty In The Examination System, 2009 20th, 21st Nov, GMIT At Davanagere.
8	“Innovative Techniques For Teaching In Technical Education”, 2006 1st-14thfeb (14 Days), BLDE,Bijapur.
7	“Advanced Cmos Technology And DSP Applications”, 2005 12th Sep, SDM, Dharwad.
6	“Arduino& Wireless Embedded Systems”, October,2016 13th To 15th, Dept Of ECE,JIT,Davanagere. NCRTCC, February, 2016, Dept Of ECE, JIT, Davanagere.
5	“Plagiarism Issues And Essential Tools For Research Write-Ups”, April, 2015, Dept Of ECE, JIT, Davanagere.
4	Workshop On “MSP 430 Processor”, 2012, Dept Of ECE, PESITM, Shimogga.
3	“National Seminar”, 2010, Dept Of Ece, GMIT, Davanagere.
2	“IEEE Conference On Emerging Devices And Smart Systems”, 3rd And 4th March, 2017, MEC, Salem.
1	“An Automatic Speaker Recognition System Using Matlab”,2012 24th ,25th August,Bheemanna Khandre Institute Of Technology,Bhalki.